

DATA SCIENCE AND MACHINE LEARNING

Day 2

Dhiraj Bashyal
AI/ML Engineer

Topics

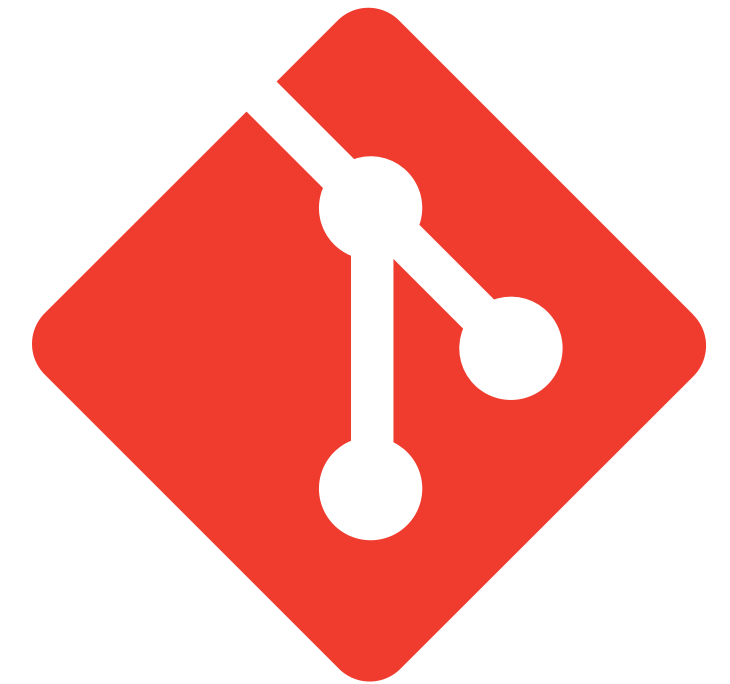
Day 2

**Git Repository Setup with
SSH Key Integration**

Git/GitHub for version control

Git is a distributed version control system

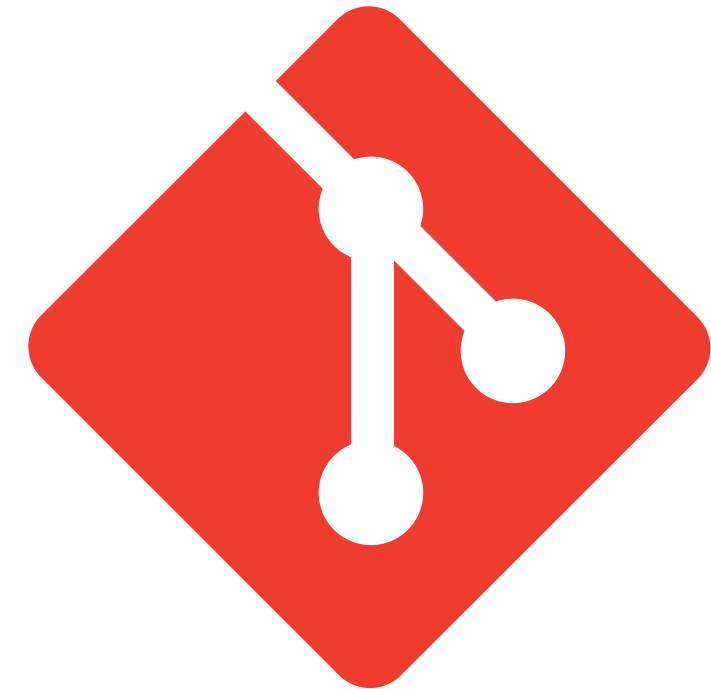
- **Why Use Git?**
 - Keeps a history of all changes.
 - Enables collaboration without overwriting others' work.
 - Provides branching for experimenting safely.
 - Easy to undo mistakes and restore previous versions.
- **Core Concepts**
 - **Repository (Repo):** Project storage with full history.
 - **Commit:** A snapshot of changes.
 - **Branch:** A parallel line of development.
 - **Merge:** Combining changes from branches.
 - **Clone:** Downloading a copy of a repo.
 - **Push/Pull:** Uploading/downloading changes.



<https://github.com/>

Git/GitHub for version control

- **git init** — Create a New Repository
 - Initializes Git in a folder.
 - Creates a hidden .git directory to start version control.
- **git clone** — Copy an Existing Repository
 - Downloads a remote repository (e.g., from GitHub) to your computer.
- **git add** — Stage Changes
 - Prepares modified files to be committed.
 - Moves changes to the staging area.
 - Example:
 - `git add file.txt`
 - `git add .` (adds all changes)
- **git commit** — Save a Snapshot
 - Records the staged changes permanently to the repository history.
 - `git commit -m "Initial commit"`
- **git push** — Upload Local Changes
 - Sends your local commits to a remote repository (GitHub, GitLab, etc.).
- **git pull** — Get Updates from Remote
 - Fetches new changes from the remote repository.
 - Automatically merges them into your local branch.

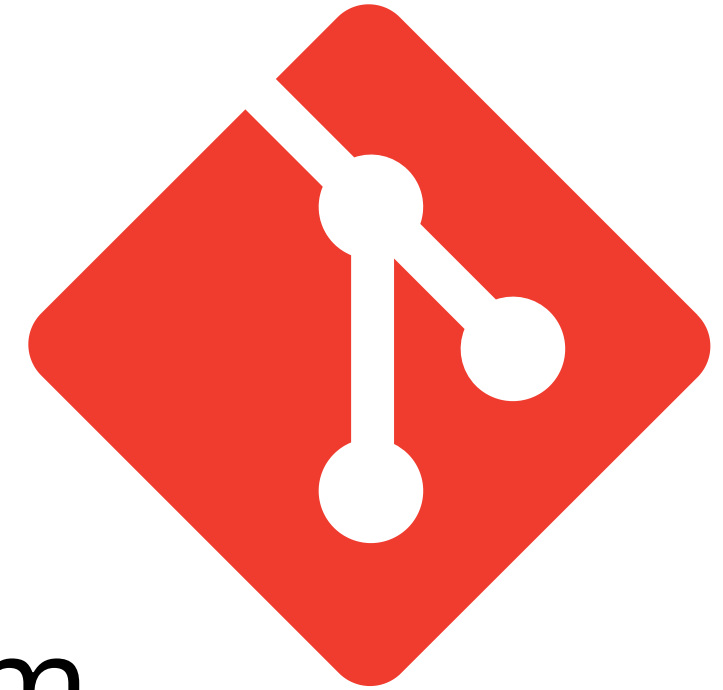


Git Playground

- <https://learngitbranching.js.org/>

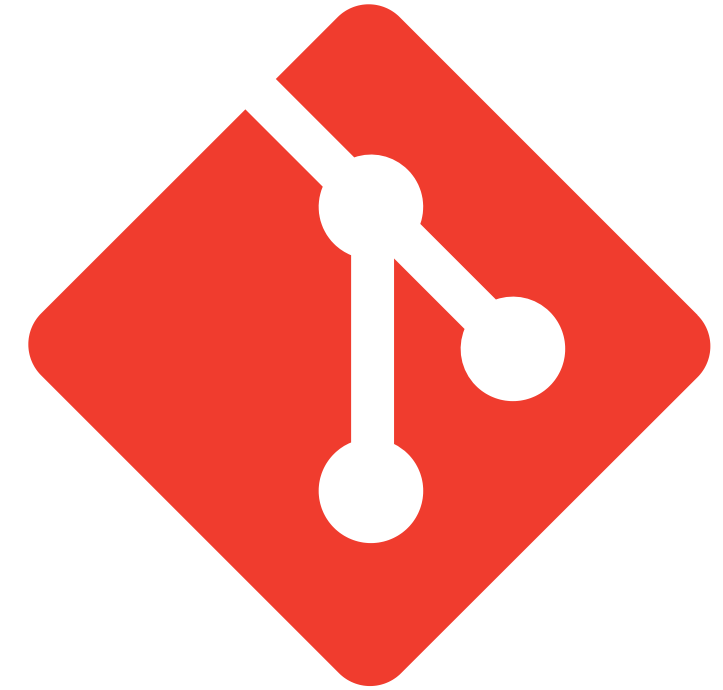
SSH Key Concept

- Public + Private key pair
- 🔑 Private stays local
- 🔓 Public added to Git platform



Check Existing Git Username and email

- `git config --global user.name`
- `git config --global user.email`



Add GitHub Email

- `git config --global user.email "your_email@example.com"`
- `git config --global user.name "Your GitHub Username"`

`git --version` # used to check the version

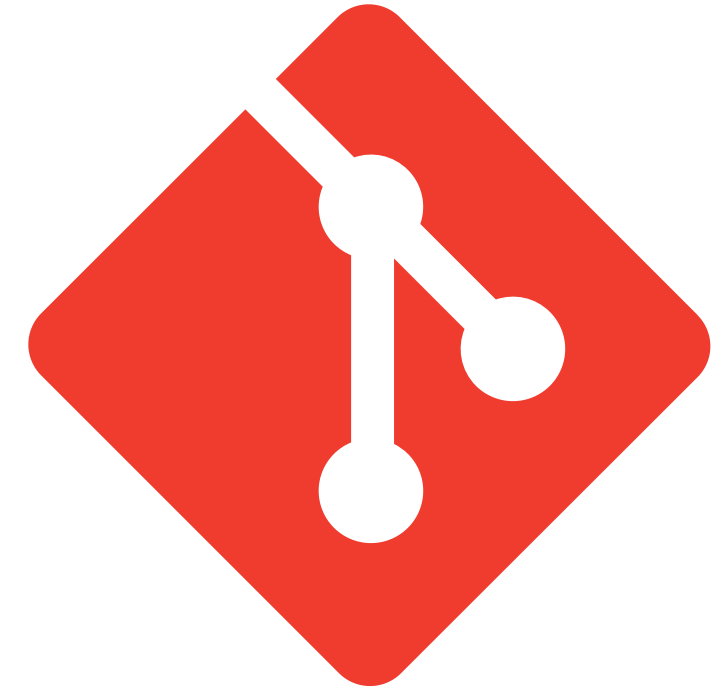
`git config --global --list`

Generate SSH Key

- `ssh-keygen -t ed25519 -C 'your
email@example.com'`

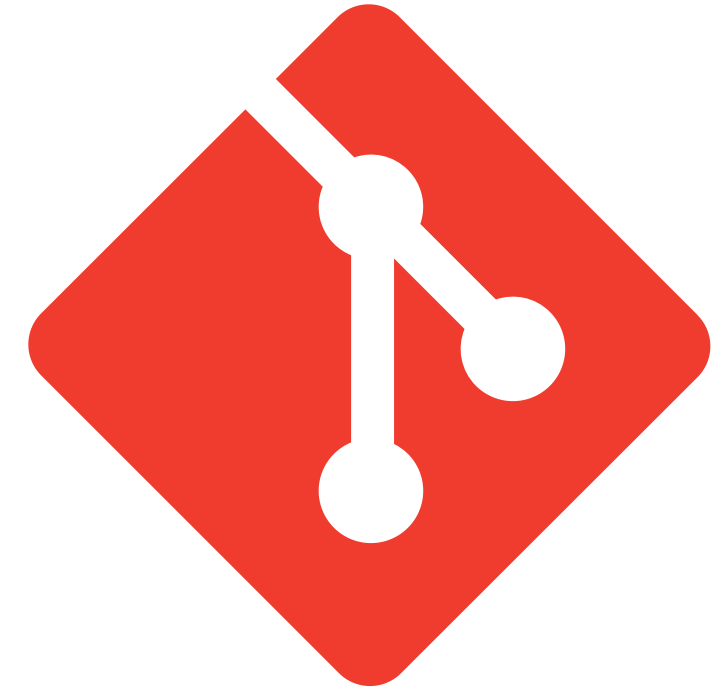
`git brach`

`git checkout -b feat/update_readme`



Start SSH Agent

- **eval \$(ssh-agent -s)**
#Start SSH agent and activate it in my terminal
- **ssh-add ~/.ssh/id_ed25519**
#adds your SSH private key to the SSH agent



Copy Public Key

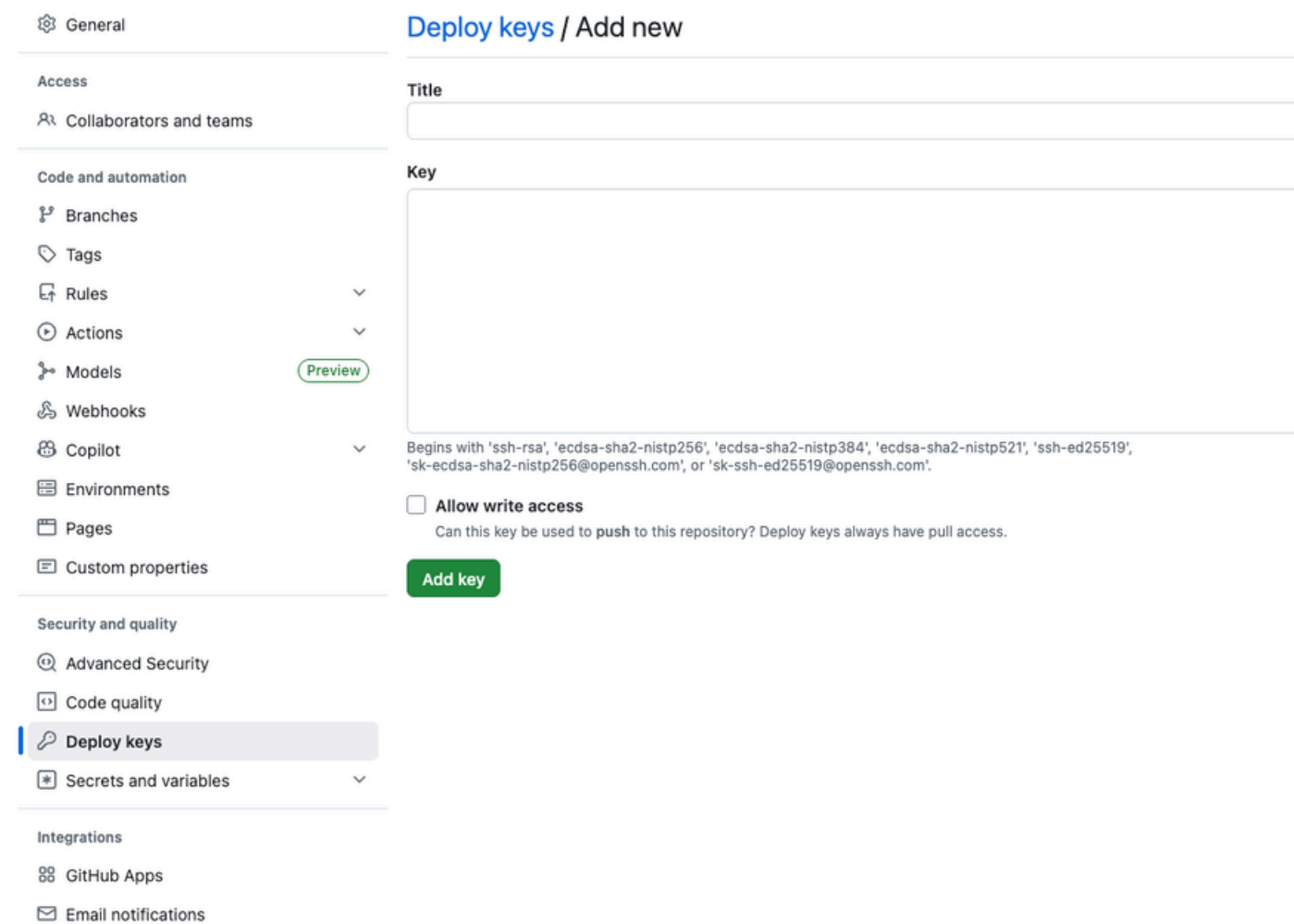
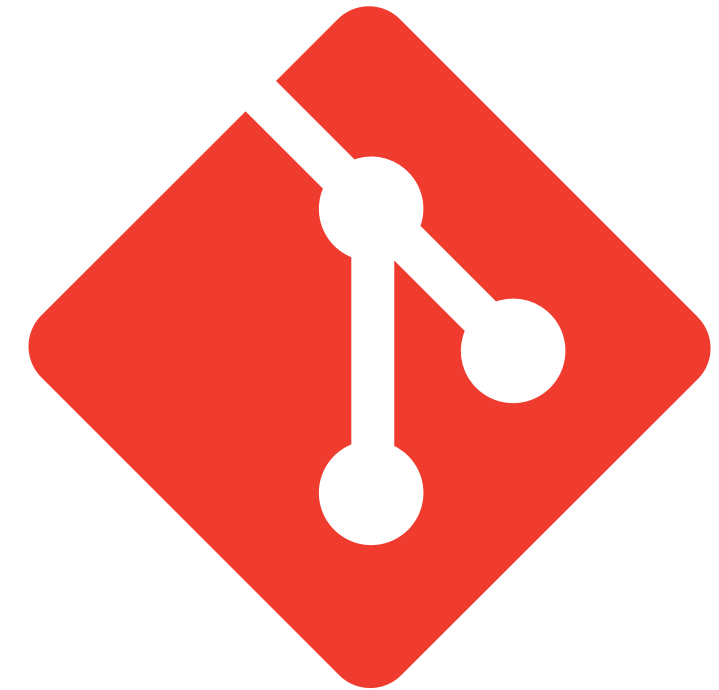
- **cat ~/.ssh/id_ed25519.pub**

simply:

- SSH key = house key 🔑
- SSH agent = security guard 🛡️
- ssh-add = giving your key to the guard so they open doors for you

Add to GitHub

Repository → Settings → Deploy keys →
Add deploy key

A screenshot of the GitHub 'Add new deploy key' interface. On the left is a sidebar with navigation links: General, Access, Collaborators and teams, Code and automation (with sub-links for Branches, Tags, Rules, Actions, Models, Webhooks, Copilot, Environments, Pages, and Custom properties), Security and quality (with sub-links for Advanced Security, Code quality, and Deploy keys), and Integrations (with sub-links for GitHub Apps and Email notifications). The 'Deploy keys' link is highlighted. The main content area is titled 'Deploy keys / Add new' and contains a 'Title' input field, a 'Key' input field with a 'Preview' button, and a checkbox for 'Allow write access' with a descriptive note. At the bottom is an 'Add key' button.

General

Deploy keys / Add new

Title

Key

Preview

Begins with 'ssh-rsa', 'ecdsa-sha2-nistp256', 'ecdsa-sha2-nistp384', 'ecdsa-sha2-nistp521', 'ssh-ed25519', 'sk-ecdsa-sha2-nistp256@openssh.com', or 'sk-ssh-ed25519@openssh.com'.

☐ Allow write access
Can this key be used to push to this repository? Deploy keys always have pull access.

Add key

Access

Collaborators and teams

Code and automation

Branches

Tags

Rules

Actions

Models

Webhooks

Copilot

Environments

Pages

Custom properties

Security and quality

Advanced Security

Code quality

Deploy keys

Secrets and variables

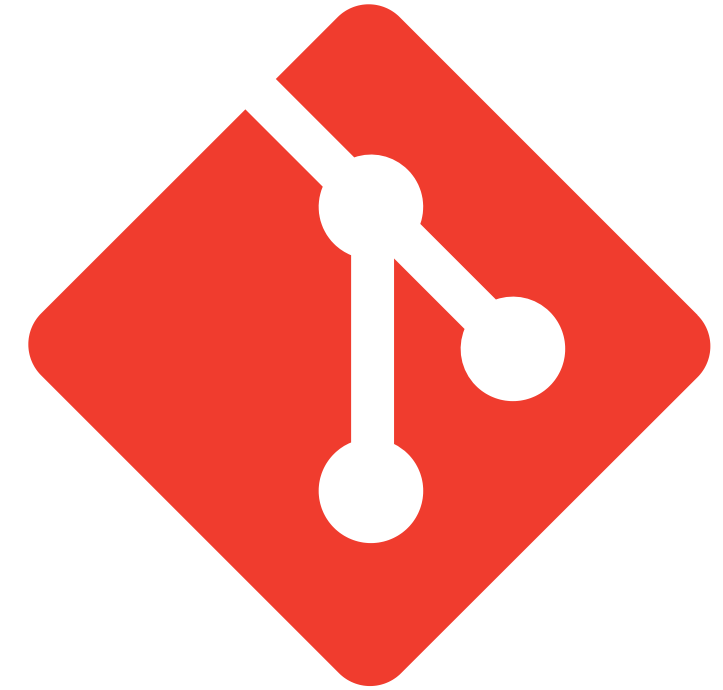
Integrations

GitHub Apps

Email notifications

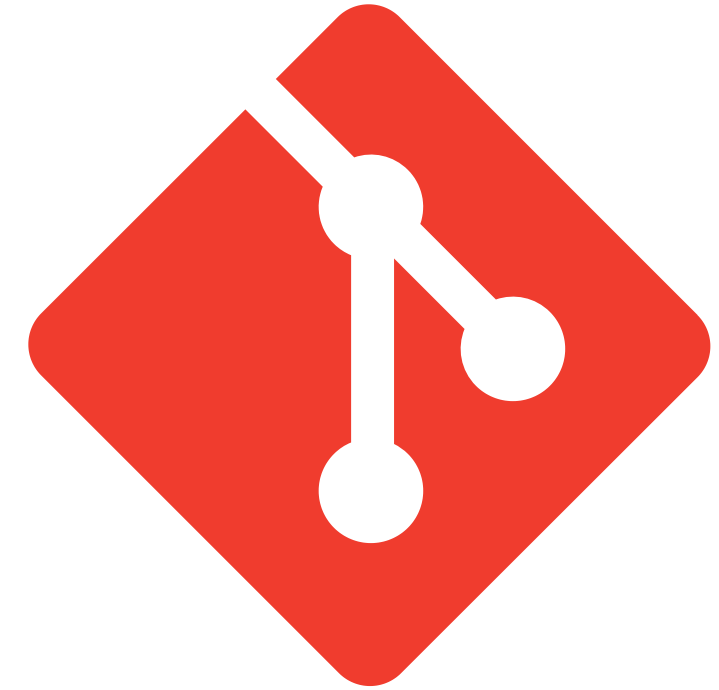
Test Connection

- `ssh -T git@github.com`



Clone Repo via SSH

```
git clone  
git@github.com:user/repo.git
```

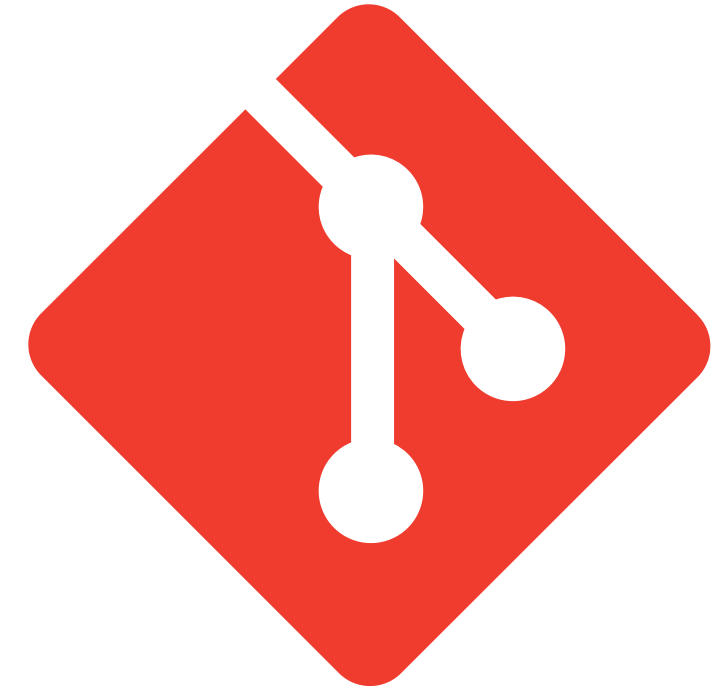


Change Remote to SSH

- `git remote set-url origin git@github.com:user/repo.git`

Clone Repo via SSH

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git clone  
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Change Remote to SSH

- `git remote set-url origin git@github.com:user/repo.git`



- git add .
- git add <file name>
- git commit -m “message”
- git status
- git push -u origin <branch name>
- git pull origin main
- git pull origin <branch name>
- git branch
- git checkout <branch name >
- git checkout -b <branch name>
- git branch -D <branch name>
#remove branch
- extra
- git stash
- git stash pop
- git stash apply

